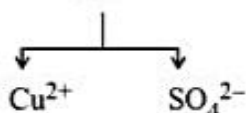
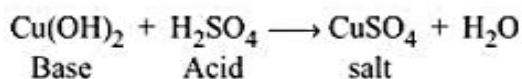


QUALITATIVE ANALYSIS

Analysis of substance is carried out in order to establish its qualitative and quantitative chemical composition. Qualitative analysis deals with identifying the components of a substance. Inorganic salts are formed due to neutralization of acids and bases. So a salt comprises of two parts a cation or basic radical contributed by the base and an anion or acidic radical contributed by an acid.

A charged atom or groups of atoms which participates in chemical reactions.



(1) Basic radical

(2) Acidic radical

*	Positive radical	–	Basic radical
*	Negative radical	–	Acid radical

- (1) **Basic radical or cation :** It is the radical having positive charge. The magnitude of charge depends upon the base in which it is in combination with OH^- ions or formally of the basic oxide. Pb^{+2} , Cu^{+2} , Al^{+3} , Fe^{+3} , Zn^{+2} , Ni^{+2} , Ca^{+2} , Ba^{+2} , Mg^{+2} , NH_4^+
- (2) **Acid radical or Anion :** It is the radical having negative charge. The magnitude of charge depends upon the acid in which it is in combination with hydrogen. CO_3^{2-} , S^{2-} , SO_4^{2-} , NO_2^- , NO_3^- , Cl^- , Br^- , I^-

Qualitative analysis : Involves the detection and identification of these radicals in salts whether single salts or double salts, or detection of radicals in mixture of salts or even radicals present in common fertilizers. Qualitative analysis is a valuable tool in industries. However in laboratories semi Micro analysis is performed.

There are separate 'procedures for detecting cations and anions, therefore qualitative analysis is studied under cation analysis and anion analysis.

Qualitative analysis deals with the identification of various constituents present in a chemical mixture. The systematic procedure involves:

1. **Preliminary tests**
 - a. Physical appearance
 - b. Dry heating test
 - c. Charcoal cavity test
 - d. Flame test
 - e. Borax bead test
2. **Indicatory tests**
 - a. Dilute acid tests
 - b. Concentrated acid test
3. **Confirmatory tests**
 - a. Physical examination
 - b. Gas evolved on heating or by adding any reagent
 - c. Analysis of acid radicals
 - d. Analysis of basic radicals

1. Preliminary tests

- (a) Physical appearance :** The physical examination of the unknown mixture involves the study of solubility colour, smell and density.

Following salts are soluble in water:

	Soluble	Insoluble
(i)	All the halides ($X^- = \text{Cl, Br, I}$) are soluble in water Except Ag, Cu, Hg, Pb,	All the CO_3^{2-} , $\text{C}_2\text{O}_4^{2-}$, S^{2-} , SO_3^{2-} are insoluble in water. Except alkyl metal
(ii)	All nitrates	Note : FeC_2O_4 is soluble in water
(iii)	All nitrites except AgNO_2	
(iv)	All sulphate are soluble in water Except Ag, Sr, Ba, Pb, Hg, Ca, Sn	
(v)	All the acetates (CH_3COO^-) are soluble in water. Except Ag, Hg	

	Experiment	Observation	Inference
(a)	Smell Take a pinch of the salt between your fingers and rub with a drop of water	Ammonical smell Vinegar like smell Smell like that of rotten eggs	NH_4^+ CH_3COO^- S^{2-}
(b)	Density	(i) Heavy (ii) Light fluffy powder	Salt of Pb^{2+} or Ba^{2+} Carbonate salts
(c)	Deliquescence	Salt absorbs moisture and becomes paste like	(i) If coloured may be $\text{Cu}(\text{NO}_3)_2$, FeCl_3 (ii) If colourless, may be $\text{Zn}(\text{NO}_3)_2$, chlorides of Zn^{2+} , Mg^{2+} etc.
(d)	Colour	Blue or Bluish green Greenish Light Green Dark brown Pink, Violet Light pink, flesh colour White	Cu^{2+} or Ni^{2+} Ni^{2+} Fe^{2+} Fe^{3+} Co^{2+} Mn^{2+} Shows the absence of Cu^{2+} , Ni^{2+} , Fe^{2+} , Fe^{3+} , Mn^{2+} , Co^{2+}

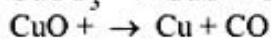
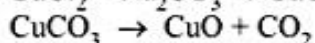
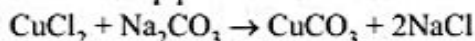
(b) Dry heating test:

On heating a small amount of mixture in a dry test tube, quite valuable information can be generated by carefully performing and noting the observations here. On heating some salts undergo decomposition thus evolving the gases or may undergo characteristic changes in the colour of residue.

Observation	Inference
(1) Gas evolved	
(a) Colourless and odourless gas CO ₂ gas - turns lime water milky	CO ₃ ²⁻
(b) Colourless gas with odour	
(i) H ₂ S gas-Smells like rotten eggs, turns lead acetate paper black.	Hydrated S ²⁻
(ii) SO ₂ gas-Characteristic suffocating smell, turns acidified potassium dichromate solution or paper green.	SO ₃ ²⁻
(iii) HCl gas- Pungent smell, white fumes with ammonia, white precipitate with silver nitrate solution.	Cl ⁻
(iv) Acetic acid vapours-Characteristic vinegar like smell.	CH ₃ COO ⁻
(v) NH ₃ gas- Characteristic smell, turns Nessler's solution brown.	NH ₄ ⁺
(c) Coloured gases-Pungent smell	
(i) NO ₂ gas-Reddish brown, turns ferrous sulphate solution black.	NO ₂ ⁻ or NO ₃ ⁻
(ii) Cl ₂ gas - Greenish yellow, turns starch iodide paper blue.	Cl ⁻
(iii) Br ₂ vapours- Reddish brown, turns starch paper orange red.	Br ⁻
(iv) I ₂ vapour- Dark violet, turns starch paper blue.	I ⁻
(2) Sublimate formed	
(a) White sublimate	NH ₄ ⁺
(b) Black sublimate accompanied by violet vapours	I ⁻
(c) Steel grey, garlic odour	As
(d) Grey sublimate	Hg
(e) Yellow sublimate	S, As ₂ S ₃
(3) Fusion test The mixture swells up into voluminous mass.	Alkali metals salts or salts containing water of crystallisation
(4) Swelling The mixture swells up into voluminous mass.	PO ₄ ³⁻ , BO ₃ ³⁻ indicated
(5) Residue	
(i) Yellow when hot, white when cold.	Zn ²⁺
(ii) Brown when hot and yellow when cold	Pb ²⁺
(iii) Original salt blue becomes white on heating	Hydrated CuSO ₄
(iv) Coloured salt becomes brown or black on heating	indicated Co ²⁺ , Fe ²⁺ , Fe ³⁺ , Cr ³⁺ , Cu ²⁺ , Ni ²⁺ , Mn ²⁺ indicated.

(c) Charcoal Cavity Test

The mixture is mixed with double of its amount of anhydrous sodium carbonate and placed in a charcoal block having a small cavity. The mass is moistened with a drop of water and heated in a reducing flame with a blow pipe. Metal salt is converted into metal via carbonate and oxide. E.g.,



The colour in the cavity bead or incrustation is observed.

S.No.	Observation	Inference
1.	Formation of metallic bead (i) Lustrous white, malleable (ii) Greyish white, marks paper (iii) White, does not mark paper (iv) Red	Ag Pb Sn Cu
2.	Incrustation with metal (i) White incrustation, brittle metal (ii) Yellow incrustation, brittle metal (iii) Yellow incrustation, malleable metal	Sb Bi Pb
3.	Incrustation without metal (i) White and yellow when hot (ii) Yellow and orange when hot (iii) Brown (iv) White (volatile, garlic odour)	ZnO, SnO BiO CdO As ₂ O ₃

(d) Flame Test

A paste of salts and concentrated HCl is taken into the lower oxidising zone and colour imparted to the flame by salts is observed. The salts of group V (Ba^{2+} , Ca^{2+} , Sr^{2+}) are identified by colours of the flame.

Colour of the flame	Inference
Greenish blue	Cu salt and BO_3^{-3}
Apple-green	Ba
Crimson-red	Sr
Brick-red	Ca
Golden-yellow	Na
Violet	K
Carmine-red	Li

Important note

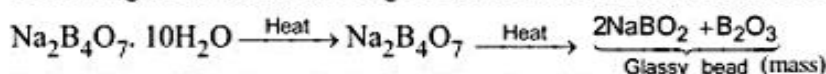
- (1) Be & Mg don't give flame test due to high ionization potential.
- (2) Colourless white salt don't possess Cu, Ni, Co, Fe, Mn, Cr etc.
- (3) White substances which swells are alum, borate and phosphate.

Sublimation Action of a Substance and Colour

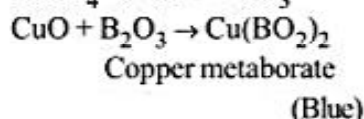
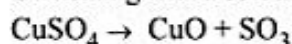
White	HgCl ₂ , Hg ₂ Cl ₂ , As ₂ O ₃ , Sb ₂ O ₃
Yellow	AlCl ₃ and NH ₃ halides
Brown	HgO, Hg(NO ₃) ₂
Blue, Black and Violet	Iodides
Black	As, Sb, Hg sulphides and iodides.

(e) Borax Bead Test

On heating borax the colourless glass bead formed consists of sodium metaborate and boric anhydride.



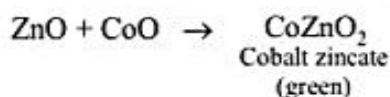
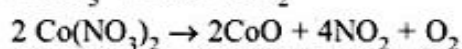
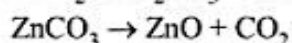
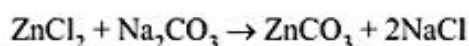
On heating with a coloured salt, the glassy bead forms a coloured metaborate in oxidising flame.



Metal	Oxidising - flame		Reducing - flame	
	Hot	Cold	Hot	Cold
Copper	Green	Blue	Colourless	Brown-red
Iron	Brown-yellow	Pale-yellow	Bottle green	Bottle green
Chromium	Green	Green	Green	Green
Cobalt	Blue	Blue	Blue	Blue
Nickel	Violet	Brown	Grey	Grey

(f) Cobalt Nitrate Charcoal Test

The mixture is mixed with double of its amount with sodium carbonate, placed in the charcoal cavity, and moistened with a drop of water and heated in an oxidising flame with the help of a blow pipe. After cooling, one or two drops of cobalt nitrate solution are added and the mass is again heated in the oxidising flame. Metal oxide combines with cobalt oxide (from cobalt nitrate) forming mixed oxide of characteristic colour.



Colour	Inference
Blue mass	Al
Green residue	Zn
Pink residue	Mg
Bluish green	Sn

IDENTIFICATION OF ACIDIC RADICALS

Group I : This group consists of radical which are detected by dilute H_2SO_4 or dilute HCl . These are (i) Carbonate, (ii) Sulphite, (iii) Sulphide, (iv) Acetate and (v) Nitrite

Group II : This group consists of radicals which are detected by concentrated H_2SO_4 . These are (i) Chloride, (ii) Bromide, (iii) Iodide, (iv) Nitrate and (v) Oxalate

Group III : The radicals which do not give any characteristic gas with dilute and concentrated H_2SO_4 . These are (i) Sulphate, (ii) Phosphate, (iii) Borate and (iv) Fluoride.

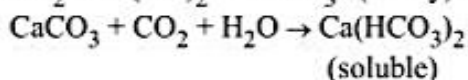
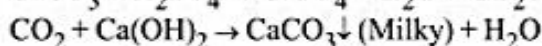
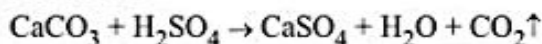
(A) **Observation of Dil. HCl / H_2SO_4 + little amount of substance on slow heating.**

(1) **CO_3^{-2} (Carbonate)**

Test/Observation/Analysis

- (i) Sharp bubbling of colourless gas (CO_2)
- (ii) Gas turns milky to lime water.
- (iii) On passing excess gas through lime water, milky colour disappears.

Reaction :

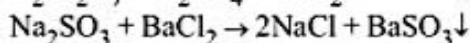
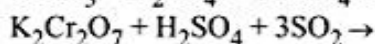
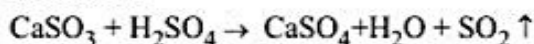


(2) **SO_3^{-2} (Sulphite)**

Test/Observation/Analysis

- (i) Colourless gas (SO_2) in which very unpleasant smell of burnt sulphur
- (ii) Gas turns green to moist acidic $K_2Cr_2O_7$ paper $K_2SO_4 + Cr(SO_4)_3$ (green) + H_2O
- (iii) Sulphite gives white ppt. with $BaCl_2$, which is soluble in dil. HCl

Reaction :

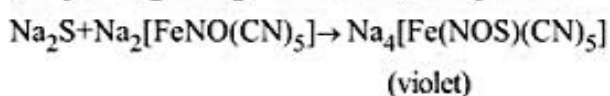
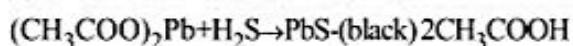
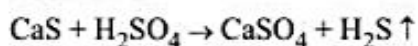


(3) S²⁻ (Sulphide)

Test/Observation/Analysis

- (i) Colourless gas with rotten egg smell (H₂S)
- (ii) Gas turns black to lead-acetate paper
- (iii) Sulphide turns violet colour to
Sodium nitroprusside solution

Reaction :

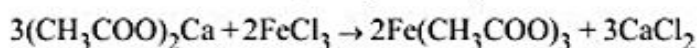
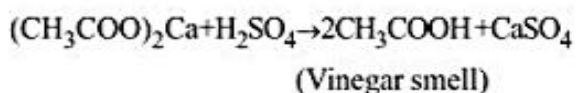


(4) CH₃COO⁻ (Acetate)

Test/Observation/Analysis

- (i) Vinegar smell, acetate may be
- (ii) Acetate gives blood red colour with neutral
FeCl₃ solution

Reaction :

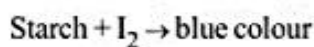
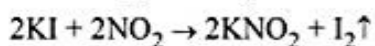
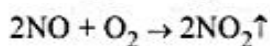
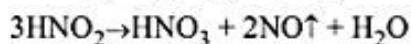
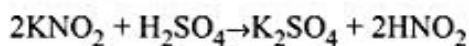


(5) NO₂⁻ (Nitrite)

Test/Observation/Analysis

- (i) Red, brown NO₂ vapour comes out. Nitrite may be
- (ii) Gas turns blue to acidic KI starch paper

Reaction :



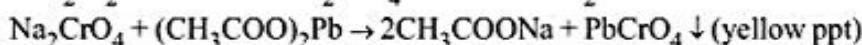
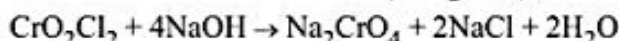
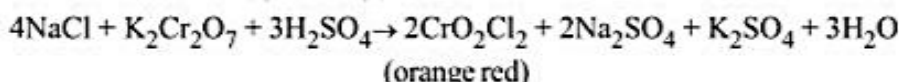
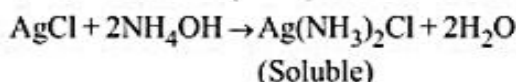
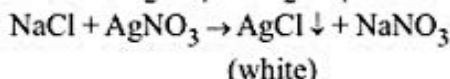
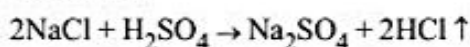
(B) Observation of Conc. H_2SO_4 + little amount of substance of slow heating

(6) Cl^- (Chloride)

Test/ observation/ Analysis

- (i) Colourless fuming gas (HCl) with fast smell
- (ii) Chloride gives white ppt. with $AgNO_3$, which is soluble in NH_4OH
- (iii) Chromyl chloride test (**v.imp.**)
 - (a) Sodium chloride when heated with $K_2Cr_2O_7$ & conc. H_2SO_4 then orange red vapour of chromyl chloride CrO_2Cl_2 comes out Hg_2Cl_2 , $PbCl_2$, $AgCl$ does not give the positive chromyl chloride test.
 - (b) This vapour when passed with $NaOH$ gives yellow solution (Na_2CrO_4)
 - (c) Acidic solution of Na_2CrO_4 gives yellow ppt. with $(CH_3COO)_2Pb$

Reaction :

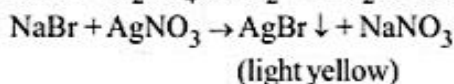
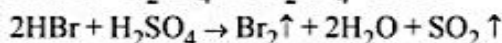
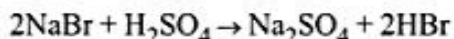


(7) Br^- (Bromide)

Test/ observation/ Analysis

- (i) Brown vapour comes out of $(Br_2)Br^-$ or NO_3^- may be
- (ii) Bromides gives light yellow ppt. with $AgNO_3$ which is partially soluble in NH_4OH .
- (iii) Brown vapour of Br_2 when passed with H_2O gives brown colouration whereas NO_2 vapour don't give any colour with H_2O

Reaction :

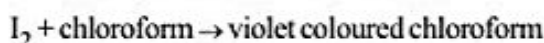
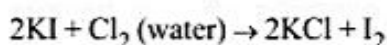
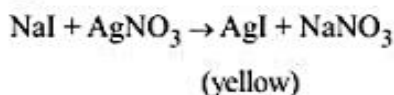
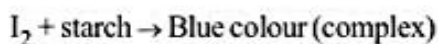
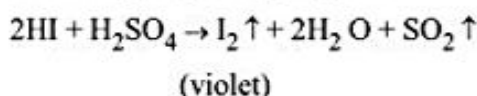
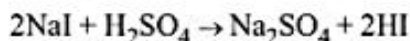


(8) I^- (Iodide)

Test/ observation/ Analysis

- (i) Dark violet fume of I_2 comes out.
- (ii) Gives blue colouration with starch.
- (iii) Iodides gives yellow ppt. with $AgNO_3$ which is insoluble in NH_4OH
- (iv) Iodine with chloroform gives violet coloured chloroform

Reaction :



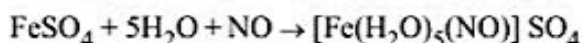
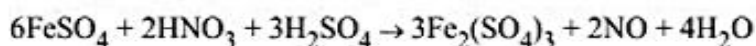
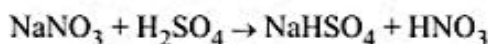
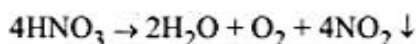
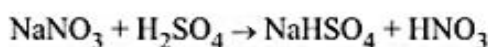
(9) NO_3^- (Nitrate)

Test/ observation/ Analysis

(i) Brown smoke comes out (NO_2)

(ii) Ring test (**v. imp.**) aq. solution of salt is mixed in fresh FeSO_4 and conc. H_2SO_4 is passed through corners of test tube, brown ring is formed. (brown ring of nitrosoferrous sulphate)

Reaction :



Solution

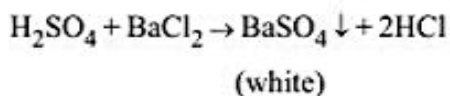
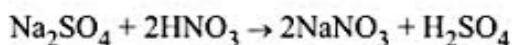
brown ring

(10) SO_4^{2-} (Sulphate)

Test/ observation/ Analysis

Small amount of substance + conc. HNO_3 mixture is heated & now adding BaCl_2 white ppt comes which is insoluble in acid or base sulphate confirmed

Reaction :



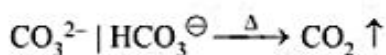
Q.4 Salt A water soluble gives pink colour with phenolphthalene and salt B will not give pink colour with phenolphthalene but both salt give colourless or odourless gas (X) on heating, gas (X) gives white turbidity with Baryta water which disappear on passing excess of gas (X). Salt A and B are.

(A) CO_3^{2-} (B) HCO_3^- (C) HSO_3^- (D) SO_3^{2-}

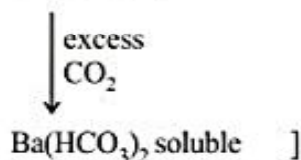
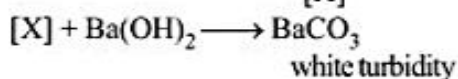
Ans. (A, B)

[Sol.] Salt A + HPh $\longrightarrow$ Pink colour
 CO_3^{2-}

Salt B + HPh $\longrightarrow$ Hot pink colour
 HCO_3^-



[X]



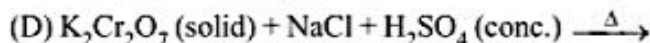
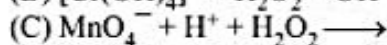
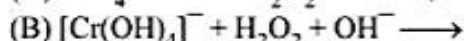
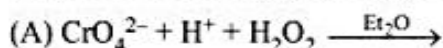
Exercise

Q.1 Borax bead test of salt (M) is performed, violet colour of the bead is obtained under oxidising flame. What is the oxidation state of the cation present in salt (M)?

(A) +7 (B) +4 (C) +2 (D) can't be predicted.

Ans. (D)

Q.2 Which of the following reaction(s) is / are not redox reaction.



Ans. (A, D)

Q.4 SO_2 gas is passed through starch iodate solution in acidic medium and the resulting solution is

(A) Salmon red coloured precipitate
(B) Red compound of unknown composition
(C) Brown colour
(D) Deep blue solution

Ans. (D)

TEST OF BASIC RADICALS

I-Group

This group includes Ag^+ , Pb^{2+} , and Hg_2^{2+} (ous). The group reagent is dilute hydrochloric acid. The radicals are precipitated as their chlorides because the solubility product of these chlorides (AgCl , PbCl_2 and Hg_2Cl_2) is less than the solubility of all other chlorides which thus remain in solution.

Radical	Test/ observation/ Analysis	Reaction
Pb^{2+}	Pb^{2+} ion gives yellow ppt. with K_2CrO_4 & KI soln. separately.	$\text{PbCl}_2 + \text{K}_2\text{CrO}_4 \rightarrow \text{PbCrO}_4(\text{yellow}) \downarrow + 2\text{KCl}$ $\text{PbCl}_2 + 2\text{KI} \rightarrow \text{PbI}_2 \downarrow (\text{yellow}) + 2\text{KCl}$
Hg_2^{2+}	Hg_2^{2+} gives black ppt. with NH_3	$\text{Hg}_2\text{Cl}_2 + 2\text{NH}_4\text{OH} \rightarrow \text{Hg}(\text{NH}_2)\text{Cl} + \text{Hg} \downarrow + \text{NH}_4\text{Cl} + 2\text{H}_2\text{O}$
Ag^+	(i) AgCl is soluble in NH_4OH (ii) Ag^+ ion gives yellow ppt. with KI	$\text{AgCl} + 2\text{NH}_4\text{OH} \rightarrow [\text{Ag}(\text{NH}_3)_2]\text{Cl} + 2\text{H}_2\text{O}$ $\text{Ag}^+ + \text{I}^- \rightarrow \text{AgI} \downarrow (\text{yellow})$

Note: PbCl_2 is soluble in hot water (but insoluble in cold water) Whereas insoluble in both AgCl & HgCl_2

II-Group

This group includes Hg^{2+} , Pb^{2+} , Bi^{3+} , Cu^{2+} , Cd^{2+} (all in IIA), As^{3+} , Sb^{3+} , Sn^{2+} and Sn^{4+} (all in IIB).

The group reagent is hydrogen sulphide in presence of HCl . These radicals are precipitated as their sulphides, whereas the sulphides of other metals remain in solution because of their high solubility product.

HgS , PbS , Bi_2S_3 and CuS are black, CdS , As_2S_3 and SnS_2 yellow, SnS brown, and Sb_2S_3 is orange.

The function of HCl is to decrease ionisation of H_2S (due to common ion effect) so that only the solubility product of sulphides of II group radical is exceeded and not that of III, IV and V group. Hence III, IV and V group radicals are not precipitated by H_2S in the presence of HCl .

PbS and CdS are precipitated only on dilution due to higher K_{sp} .

IIA and IIB group sulphides are separated by yellow ammonium sulphide $(\text{NH}_4)_2\text{S}_x$ in which IIA group sulphides are insoluble whereas that of IIB are soluble forming thioalts.

Radical	Test/ observation/ Analysis	Reaction
Hg⁺²	Adding Hg ⁺² ion white ppt. obtained which turns black	$2\text{Hg}^{+2} + \text{SnCl}_2 \rightarrow \text{Sn}^{+4} + \text{Hg}_2\text{Cl}_2 \downarrow$ (white) $\text{HgCl}_2 + \text{SnCl}_2 \rightarrow \text{SnCl}_4 + 2\text{Hg} \downarrow$ (black)
Pb⁺²	(i) In solution, Pb ⁺² gives white ppt. with H ₂ SO ₄ (ii) In solution Pb ⁺² ion gives yellow ppt. with K ₂ CrO ₄ & KI	$\text{Pb}^{+2} + \text{H}_2\text{SO}_4 \rightarrow \text{PbSO}_4 \downarrow + 2\text{H}^+$ (white) $\text{Pb}^{+2} + \text{Cr}_2\text{O}_4^{2-} \rightarrow \text{PbCrO}_4 \downarrow \text{ (yellow)}$ $\text{Pb}^{+2} + 2\text{I}^- \rightarrow \text{PbI}_2 \downarrow \text{ (yellow)}$
Cu⁺²	(i) These ion gives dark blue colour with excess NH ₄ OH (ii) Cu ⁺² ion gives chocolate colour with K ₄ Fe(CN) ₆	$\text{Cu}^{+2} + \text{NH}_4\text{OH} \rightarrow [\text{Cu}(\text{NH}_3)_4]^{+2} + \text{H}_2\text{O}$ (dark blue colour) $2\text{Cu}^{+2} + \text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Cu}_2[\text{Fe}(\text{CN})_6] \downarrow + 4\text{K}^+$ (chocolate or red brown ppt.)
Bi⁺³	Bi ⁺³ ion gives white ppt. while adding water in HCl soln.	$\text{BiCl}_3 + \text{H}_2\text{O} \rightarrow \text{BiOCl} \downarrow + 2\text{HCl}$ (white bismuth oxychloride) $\text{BiCl}_3 + 3\text{Na}_2\text{SnO}_2 + 6\text{NaOH} \rightarrow$ (sodium stanite) $2\text{Bi} \downarrow + 3\text{Na}_2\text{SnO}_3 + 6\text{NaCl} + 3\text{H}_2\text{O}$ (black sodium stanate)
Cd⁺²	(i) The yellow precipitate is dissolved in 50% HNO ₃ . To the resulting solution, NH ₄ OH is added slowly. A white ppt. appears which dissolve in excess of NH ₄ OH. (ii) When H ₂ S gas is passed in this solution a yellow ppt. appears	$3\text{Cds} + 8\text{HNO}_3 \rightarrow 3\text{Cd}(\text{NO}_3)_2 + 4\text{H}_2\text{O} + 2\text{NO} + 3\text{S}$ (50%) $\text{Cd}(\text{NO}_3)_2 + 2\text{NH}_4\text{OH} \rightarrow 2\text{NH}_4\text{NO}_3 + \text{Cd}(\text{OH})_2 \downarrow$ white ppt. $\text{Cd}(\text{OH})_2 + 2\text{NH}_4\text{OH} + 2\text{NH}_4\text{NO}_3 \rightarrow [\text{Cd}(\text{NH}_3)_4](\text{NO}_3)_2\text{aq.} + 4\text{H}_2\text{O}$ $[\text{Cd}(\text{NH}_3)_4](\text{NO}_3)_2 + \text{H}_2\text{S} \rightarrow \text{CdS} \downarrow + 2\text{NH}_4\text{NO}_3 + 2\text{NH}_3$ (yellow ppt.)

III group

This group includes Fe³⁺, Al³⁺, and Cr³⁺. The group reagent is NH₄OH and in presence of NH₄Cl, the radicals are precipitated as their hydroxides. The function of NH₄Cl is to suppress the ionisation of NH₄OH so that only the IIIrd group radicals are precipitated, because the solubility product of IIIrd group hydroxides is less than that of IV and VI group hydroxides.

Excess of NH₄Cl should be added, or else manganese will be precipitated in III group.

Radical	Test/ observation/ Analysis	Reaction
Fe⁺³, Cr⁺³ & Al⁺³	These ion precipitates in the form of hydroxide on adding NH ₄ Cl & NH ₄ OH	$\text{Fe}^{+3} + 3\text{OH}^- \rightarrow \text{Fe}(\text{OH})_3 \text{ (red ppt.)}$ $\text{Cr}^{+3} + 3\text{OH}^- \rightarrow \text{Cr}(\text{OH})_3 \text{ (green ppt.)}$ $\text{Al}^{+3} + 3\text{OH}^- \rightarrow \text{Al}(\text{OH})_3 \text{ (white ppt.)}$

Note : In the analysis of III group, some drops of conc. HNO₃ are also added before oxidising Fe⁺² to Fe⁺³.

Al³⁺	White ppt. of Al(OH) ₃ is soluble in NaOH	Imp. Al(OH) ₃ + NaOH → NaAlO ₂ + 2H ₂ O (sodium metaaluminate)
Cr³⁺	ppt. of Cr(OH) ₃ is soluble in NaOH + Br ₂ water soln. in this soln. when BaCl ₂ is added yellow	Br ₂ + H ₂ O → 2HBr + O 2Cr(OH) ₃ + 4NaOH + 3O → 2Na ₂ CrO ₄ + 5H ₂ O ppt. is obtained Na ₂ CrO ₄ + BaCl ₂ → BaCrO ₄ ↓ (yellow ppt.) + 2NaCl
Fe³⁺	(i) (a) Brown ppt. of Fe(OH) ₃ is soluble in HCl (b) When KCNS is added in this soln. soln. blood red colouration is obtained (ii) In this soln., on adding K ₄ [Fe(CN) ₆], prussian blue colour is obtained	Fe(OH) ₃ + 3HCl → FeCl ₃ + 3H ₂ O FeCl ₃ + 3KCNS → Fe(CNS) ₃ + 3KCl (ferric thiocyanate) (blood red) 4FeCl ₃ + 3K ₄ [Fe(CN) ₆] → Fe ₄ [Fe(CN) ₆] ₃ + 12KCl (ferric ferrocyanide prussian blue)

IV group

This group includes Co²⁺, Ni²⁺, Zn²⁺ and Mn²⁺. The group reagent is hydrogen sulphide in ammonical solution. Radicals are precipitated as sulphides which are insoluble in NH₄OH.

CoS, NiS - Black

MnS - Buff

ZnS-White

The function of ammonium hydroxide is to increase the ionisation of H₂S.

Thus, an excess of sulphide ions will be available and hence the ionic product of IV group sulphides exceeds their solubility product and precipitates will be obtained.

Radical	Test/ observation/ Analysis	Reaction
Zn²⁺, Mn²⁺ Co²⁺, Ni²⁺ Co²⁺, Ni²⁺ Zn²⁺ Mn²⁺ Ni²⁺	These ions in presence of NH ₄ OH precipitate on passing H ₂ S. Black (CoS, NiS) ppt., (soluble in aqua-regia) White (ZnS) (soluble in HCl) Pink or buff (MnS), soluble in HCl In presence of NH ₄ OH, Ni salt on reaction with dimethyl glyoxime (DMG) turns red ppt. of nickel dimethyl glyoxime → Nickel dimethyl glyoxime (red ppt)	MCl ₂ + H ₂ S → MS ↓ + 2HCl V. Imp. CH ₃ -C=NOH + NiCl ₂ + 2NH ₄ OH CH ₃ -C=NOH → (C ₄ H ₇ N ₂ O ₂) ₂ Ni ↓ + 2NH ₄ Cl + 2H ₂ O
Co²⁺	Cobalt salt turns blue colouration with NH ₄ CNS	CoCl ₂ + 4NH ₄ CNS → (NH ₄) ₂ [Co(CNS) ₄] + 2NH ₄ Cl (ammonium cobalt thiocyanate) (blue colour)
Zn²⁺	In solution, Zn ²⁺ ion turns white ppt. with NaOH which is soluble in excess NaOH	V. Imp Zn ²⁺ + 2NaOH → Zn(OH) ₂ ↓ (white) + 2Na Zn(OH) ₂ + 2NaOH → Na ₂ ZnO ₂ + 2H ₂ O
Mn²⁺	(a) Mn ²⁺ ion gives pink ppt. with NaOH (b) On heating turns black or brown	V. Imp Mn ²⁺ + 2NaOH → Mn(OH) ₂ ↓ + 2Na (Pink) Mn(OH) ₂ + O $\xrightarrow{\Delta}$ MnO ₂ + H ₂ O (brown and black)

V group

This includes Ba^{2+} , Sr^{2+} and Ca^{2+} . The group reagent is ammonium carbonate in the presence of NH_4Cl and NH_4OH . These are precipitated as carbonates which are insoluble in NH_4OH .

The function of ammonium chloride is to suppress the ionisation of NH_4OH and $(\text{NH}_4)_2\text{CO}_3$ and thus check the precipitation of $\text{Mg}(\text{OH})_2$ (along with V group carbonates) because the solubility product of $\text{Mg}(\text{OH})_2$ and MgCO_3 is high. Further NH_4Cl should also not be added in excess, as the high concentration of NH_4^+ ions will decrease the ionisation of $(\text{NH}_4)_2\text{CO}_3$ to such an extent that sufficient CO_3^{2-} ions may not be present and carbonates of this group of metals may not precipitate.

Radical	Test/ observation/ Analysis	Reaction
Ba^{+2} , Sr^{+2} , Ca^{+2}	On adding $(\text{NH}_4)_2\text{CO}_3$, these precipitates in the form of carbonates. soluble in CH_3COOH	$\text{M}^{+2} + (\text{NH}_4)_2\text{CO}_3 \rightarrow \text{MCO}_3 + 2\text{NH}_4^+$ $\text{BaCO}_3, \text{CaCO}_3, \text{SrCO}_3$ (white)
Ba^{+2}	Gives Ba^{+2} ion in solution (i) Yellow ppt. with K_2CrO_4 (ii) white ppt. with $(\text{NH}_4)_2\text{SO}_4$ (iii) white ppt. with $(\text{NH}_4)_2\text{C}_2\text{O}_4$	$\text{Ba}^{+2} + \text{K}_2\text{CrO}_4 \rightarrow \text{BaCrO}_4 \downarrow$ (yellow) + 2K^+ $\text{Ba}^{+2} + (\text{NH}_4)_2\text{SO}_4 \rightarrow \text{BaSO}_4 \downarrow$ (white) + 2NH_4^+ $\text{Ba}^{+2} + (\text{NH}_4)_2\text{C}_2\text{O}_4 \rightarrow \text{BaC}_2\text{O}_4 \downarrow$ (white) + 2NH_4^+
Sr^{+2}	Sr^{+2} ion with $(\text{NH}_4)_2\text{SO}_4$ gives white precipitate	$\text{Sr}^{+2}(\text{NH}_4)_2\text{SO}_4 \rightarrow \text{SrSO}_4 \downarrow + 2\text{NH}_4^+$ (white ppt.)
Ca^{+2}	Ca^{+2} ion gives white ppt. only with $(\text{NH}_4)_2\text{C}_2\text{O}_4$	$\text{Ca}^{+2} + (\text{NH}_4)_2\text{C}_2\text{O}_4 \rightarrow \text{CaC}_2\text{O}_4 \downarrow + 2\text{NH}_4^+$ (white) $\text{Sr}^{+2} + (\text{NH}_4)_2\text{C}_2\text{O}_4 \rightarrow \text{SrC}_2\text{O}_4 \downarrow + 2\text{NH}_4^+$ (white ppt.)

Note : The order of that is same as above Ba^{+2} , Sr^{+2} , Ca^{+2}

VI Group

Radical	Test/ observation/ Analysis	Reaction
Mg^{+2} Zero group/ NH_4^+	Mg^{+2} ion gives white ppt. with NH_4OH & $(\text{NH}_4)_2\text{HPO}_4$ (i) (a) All ammonium salts on reacting with base like (NaOH) , gives smell of NH_3 (b) Gas evolved (NH_3) gives white fume with HCl (c) On passing NH_3 in $\text{Hg}_2(\text{NO}_3)_2$, black colour is obtained (b) Brown ppt. is obtained with nessler's reagent	$\text{Mg}^{+2} + (\text{NH}_4)_2\text{HPO}_4 + \text{NH}_4\text{OH} \rightarrow \text{MgNH}_4\text{PO}_4 \downarrow$ (white) + $2\text{NH}_4^+ + \text{H}_2\text{O}$ (a) $\text{NH}_4\text{Cl} + \text{NaOH} \rightarrow \text{NaCl} + \text{NH}_3 \uparrow + \text{H}_2\text{O}$ (b) $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl} \uparrow$ (white fume) (c) $\text{Hg}_2(\text{NO}_3)_2 + 2\text{NH}_3 \rightarrow \text{Hg} + \text{Hg}(\text{NH}_2)\text{NO}_3 + \text{NH}_4\text{NO}_3$ (black) $\rightarrow$ (d) $2\text{K}_2\text{HgI}_4 + 4\text{KOH} + \text{NH}_4\text{Cl} \rightarrow$ (Nessler's reagent) $\begin{array}{c} \text{NH}_2 \\ \\ \text{Hg} \\ \\ \text{O} \\ \\ \text{Hg} \\ \\ \text{I} \end{array} + 7\text{KI} + \text{KCl} \cdot 3\text{H}_2\text{O}$ (Iodide solution black or brown ppt.) millions base

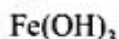
Illustration

Q.1 During group analysis, reddish brown ppt. is observed in group-III. What is the oxidation state of the metal present in the above precipitate.

- (A) +2 (B) +3 (C) both +2 and +3 (D) Can't be predicted

Ans. (B)

[Sol.] +3



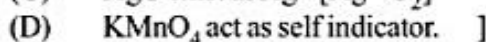
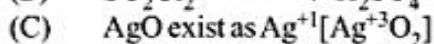
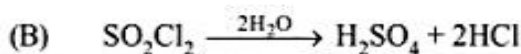
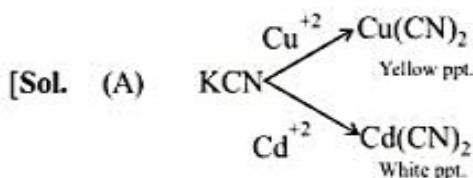
Reddish brown

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Q.2 Which of the following statement is correct.

- (A) Using KCN solution Cu^{2+} and Cd^{2+} ion can not be distinguished
(B) SO_2Cl_2 on hydrolysis produces two molecules of HCl and Caro's acid.
(C) The oxidation state of Ag in AgO is +2.
(D) No indicator is required in the estimation of KMnO_4 using standard oxalic acid solution

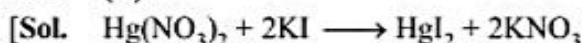
Ans. (D)



Q.3 KI is added in excess into $\text{Hg}(\text{NO}_3)_2$ solution. The observation is

- (A) Yellow ppt. of HgI_2 (B) Scarlet red ppt. of HgI_2
(C) Colourless solution of $[\text{HgI}_4]^{2-}$ (D) None of these

Ans. (C)



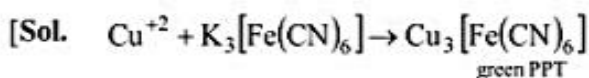
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Q.4 Unknown solution of salt 'A' $\xrightarrow{\text{K}_3[\text{Fe}(\text{CN})_6]}$ green ppt is obtained.

Which of the following radicals will be confirmed.

- (A) Ni^{2+} (B) Cu^{2+} (C) $\text{S}_2\text{O}_3^{2-}$ (D) SO_3^{2-}

Ans. (B)



Exercise

Q.1 Select correct statement(s)

- (I) When excess FeCl_3 solution is added to $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution in addition to $\text{Fe}^{\text{III}}[\text{Fe}^{\text{II}}(\text{CN})_6]^-$, $\text{Fe}^{\text{II}}[\text{Fe}^{\text{III}}(\text{CN})_6]^-$ is also formed due to side redox reaction
 (II) When FeCl_2 is added to $\text{K}_3[\text{Fe}(\text{CN})_6]$ solution, in addition to $\text{Fe}^{\text{II}}[\text{Fe}^{\text{III}}(\text{CN})_6]^-$, $\text{Fe}^{\text{III}}[\text{Fe}^{\text{II}}(\text{CN})_6]^-$ is also formed due to side redox reaction.
 (III) $\text{Fe}^{\text{III}}[\text{Fe}^{\text{II}}(\text{CN})_6]^-$ is paramagnetic while $\text{Fe}^{\text{II}}[\text{Fe}^{\text{III}}(\text{CN})_6]^-$ is diamagnetic
 (IV) $\text{Fe}^{\text{III}}[\text{Fe}^{\text{II}}(\text{CN})_6]^-$ is diamagnetic while $\text{Fe}^{\text{II}}[\text{Fe}^{\text{III}}(\text{CN})_6]^-$ is paramagnetic
 (A) I, II (B) III, IV (C) All (D) None

Ans. (A)

Q.2 Which of the following compound is/are partially soluble or insoluble in NH_4OH solution

- (1) $\text{Fe}(\text{OH})_3$ (2) Ag_2CrO_4 (3) $\text{Al}(\text{OH})_3$ (4) Ag_2CO_3 (5) $\text{Ni}(\text{OH})_2$
 (A) 1, 3, 5 (B) 2, 3, 4 (C) 1, 3 (D) 2, 3, 5

Ans. (C)

Q.3 The sulphide which is insoluble in both ammonium sulphide and HNO_3 is

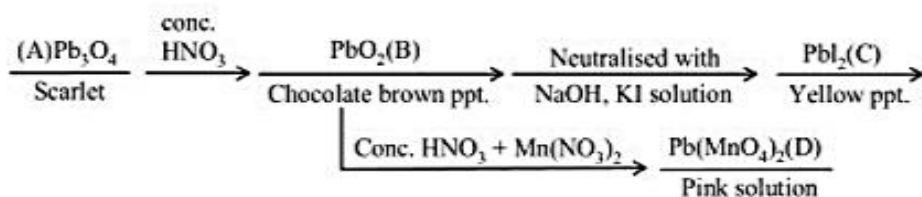
- (A) Bi_2S_3 (B) CuS (C) FeS (D) HgS

Ans. (D)

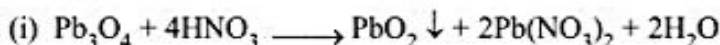
SOLVED EXAMPLE

Q.1 A scarlet compound A is treated with concentrated HNO_3 to give a chocolate-brown precipitate B. The precipitate is filtered and the filtrate is neutralised with NaOH . Addition of KI to the resulting solution gives a yellow precipitate C. The precipitate B on warming with conc. HNO_3 in the presence of $\text{Mn}(\text{NO}_3)_2$ produces a pink-coloured solution due to the formation of D. Identify A, B, C, and D. Also, write the reaction sequence.

Sol. Since yellow precipitate (C) is obtained with KI and the pink-coloured solution is formed with concentrated HNO_3 and $\text{Mn}(\text{NO}_3)_2$, the compound should be of Pb. We know that Pb_3O_4 is a scarlet compound.



The equations are as follows:



Q.2 A certain salt (X) gives the following tests:

(i) Its aqueous solution is alkaline to litmus.

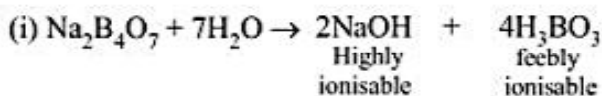
(ii) On strong heating, it swells to give a glassy material.

(iii) When concentrated sulphuric acid is added to a hot concentrated solution of (X), white crystals of an acid separate out.

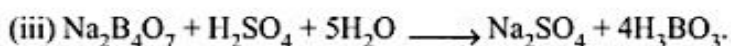
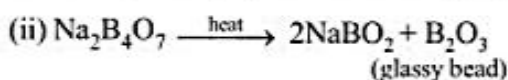
Identify (X) and write down the chemical equations for reactions at step (i), (ii), and (iii).

Sol. Since the aqueous solution of salt (X) is alkaline to litmus, it should be sodium or potassium salt. It swells to give a glassy material on heating, the salt (X) should be borax, that is, $\text{Na}_2\text{B}_4\text{O}_7$ or $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$. It is further confirmed by the reaction of borax and concentrated H_2SO_4 as it gives boric acid which is a white crystalline compound. Boric acid is a weak acid.

The reactions are as follows:



Since NaOH is highly ionisable and H_3BO_3 is feebly ionisable, the solution is alkaline.

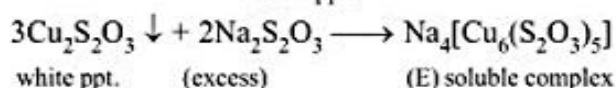
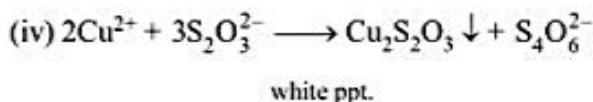
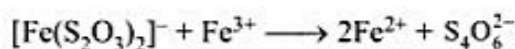
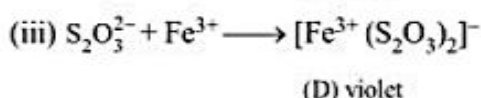
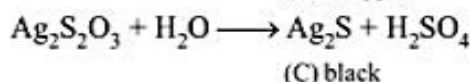
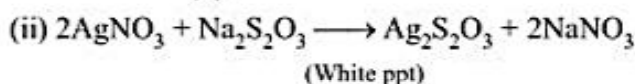
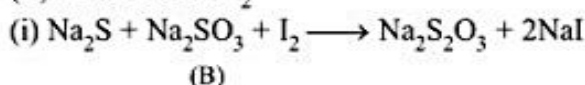


We know that H_3BO_3 is a white crystalline compound and is a weak acid.

Q.3

- (i) An inorganic compound (A) is formed on passing a gas (B) through a conc. liquor containing Na_2S and sodium sulphite.
 - (ii) On adding (A) into a dilute solution of silver nitrate a white precipitate appears which quickly changes into a black coloured compound (C).
 - (iii) On adding two or three drops of FeCl_3 into the excess of solution (A) a violet coloured compound (D) is formed. This colour disappears quickly.
 - (iv) On adding a solution of (A) into the solution of cupric chloride, a white precipitate is first formed which dissolves on adding excess of (A) forming a compound (E).
- Identify (A) to (E) and give chemical equations for the reactions at steps (i) to (iv).

Sol. The reactions indicate that the compound (A) is sodium thiosulphate. It is formed in step (i) by passing gas (B) which is either I_2 .



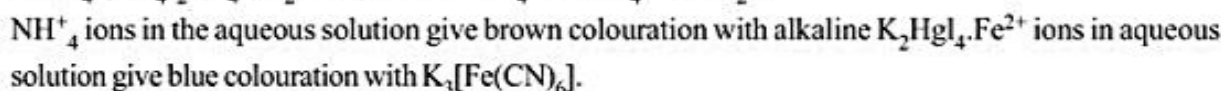
Q.4

- (i) A blue coloured compound (A) on heating gives two of the products (B) & (C).
 - (ii) A metal (D) is deposited on passing hydrogen through heated (B).
 - (iii) The solution of (B) in HCl on treatment with the $\text{K}_4[\text{Fe}(\text{CN})_6]$ gives a chocolate brown coloured precipitate of compound (E).
 - (iv) (C) turns lime water milky which disappears on continuous passage of (C) forming a compound (F).
- Identify (A) to (F) and give chemical equations for the reactions at step (i) to (iv).

Sol. $\text{A} = 2\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$, $\text{B} = \text{CuO}$, $\text{C} = \text{CO}_2$, $\text{D} = \text{Cu}$, $\text{E} = \text{Cu}_2[\text{Fe}(\text{CN})_6]$, $\text{F} = \text{Ca}(\text{HCO}_3)_2$.
Azurite

Q.5 A light bluish green crystalline compound responds to the following tests:

- (i) Its aqueous solution gives a brown precipitate or colouration with alkaline $\text{K}_2[\text{HgI}_4]$ solution.
 - (ii) Its aqueous solution gives a blue colour with $\text{K}_3[\text{Fe}(\text{CN})_6]$ solution.
 - (iii) Its solution in hydrochloric acid gives a white precipitate with BaCl_2 solution.
- Identify the ions present and suggest the formula of the compound.

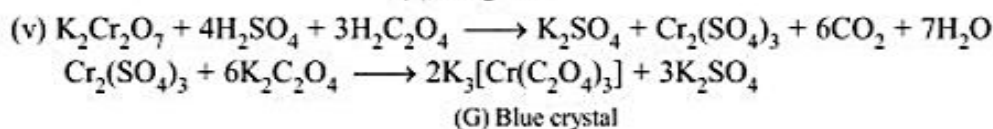
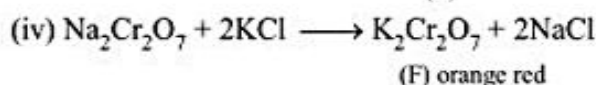
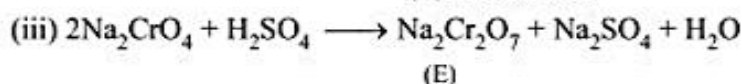
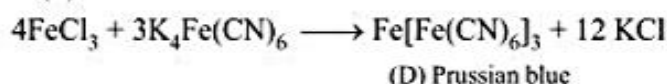
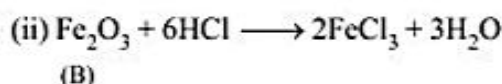
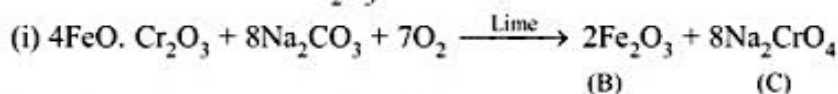


2SO_4^{2-} ions in aqueous solution (dilute HCl) give a white precipitate with BaCl_2 solution.

- (i) An ore (A) on roasting with sodium carbonate and lime in the presence of air gives two compounds, (B) and (C).
- (ii) The solution of (B) in conc. HCl on treatment with potassium ferrocyanide gives a blue colour or precipitate of compound (D).
- (iii) The aqueous solution of (C) on treatment with conc. H_2SO_4 gives a orange coloured compound (E).
- (iv) Compound (E) when treated with KCl gives an orange red compound (F) which is used as an oxidising reagents.
- (v) The solution of (F) on treatment with oxalic acid and then with an excess of potassium oxalate gives blue crystals of compound (G).

Identify (A) to (G) and give balanced chemical equations for reactions at step (i) to (v).

Sol. The ore is chromite $\text{FeO} \cdot \text{Cr}_2\text{O}_3$.



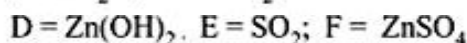
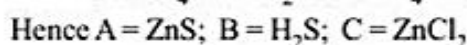
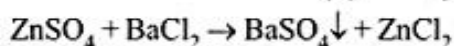
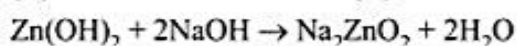
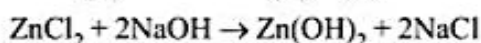
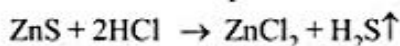
Q.7 A white substance (A) on heating with excess of dilute HCl gave an offensive-smelling gas (B) and a solution (C). Solution (C) on treatment with aqueous ammonia did not give any precipitate but on treatment with NaOH solution gave a precipitate (D), which dissolves in excess of NaOH solution. (A) on strong heating in air gave a strong-smelling gas (E) and a solid (F). Solid (F) dissolved completely in HCl and the solution gave a precipitate with BaCl_2 in acid solution. Identify (A) to (F) and write the chemical equations for the various reactions involved.

Sol. Since solution C gives precipitate with NaOH solution which is soluble in excess of NaOH, the cation

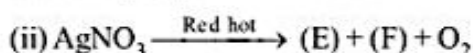
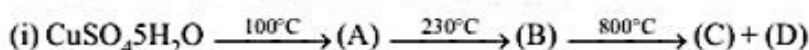
should be an amphoteric metal like Zn or Al. Again, solid F is soluble in HCl and gives a white precipitate with BaCl_2 . Therefore, the anion must be SO_4^{2-} ion.

Substance A gives an offensive-smelling gas and, thus, the compound A may be ZnS or Al_2S_3 . But, Al_2S_3 on heating in air does not form $\text{Al}_2(\text{SO}_4)_3$.

$\therefore$ It is concluded that the compound A is ZnS . The chemical reactions are as follows:



Q.8 Complete the following by identifying (A) to (F).



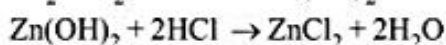
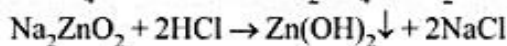
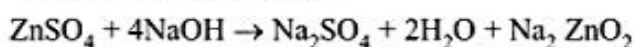
Sol. $\text{A} = \text{CuSO}_4 \cdot \text{H}_2\text{O}$, $\text{B} = \text{CuSO}_4$, $\text{C} = \text{CuO}$, $\text{D} = \text{SO}_3$, $\text{E} = \text{Ag}$, $\text{F} = \text{NO}_2$

Q.9 An unknown solid mixture contains one or two of the following: CaCO_3 , BaCl_2 , AgNO_3 , Na_2SO_4 , ZnSO_4 , and NaOH . The mixture is completely soluble in water and the solution gives pink colour with phenolphthalein. When 0.1 N HCl solution is gradually added to the above solution, a precipitate is produced which dissolves with further addition of the acid. What is /are present in the solid? Give equations to explain the appearance of the precipitate and its dissolution.

Sol. Since the mixture is completely soluble in water, the presence of CaCO_3 is ruled out. Further, the solution of the mixture gives pink colour with phenolphthalein; therefore, NaOH is present in the mixture. Again, the precipitate is dissolved by adding HCl. This rules out the possibility of BaCl_2 , AgNO_3 , or Na_2SO_4 . It is because if BaCl_2 is there, the precipitate of Ba(OH)_2 should be formed in the presence of NaOH whereas the mixture is completely soluble. If AgNO_3 is there, the precipitate of AgCl should be formed with HCl. There is no possibility to form precipitate with Na_2SO_4 .

Now ZnSO_4 may be the second compound of the mixture. Actually, NaOH is in excess in the mixture. Therefore, sodium zincate is formed in the solution which is soluble in water. On adding HCl in Na_2ZnO_2 , a precipitate of Zn(OH)_2 forms which dissolves in HCl to form ZnCl_2 .

The reactions are as follows:



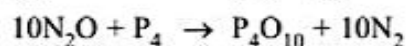
Hence, the mixture contains NaOH and ZnSO_4 .

Q.10 A colourless inorganic salt (A) decomposes completely at about 250°C to give only two products, (B) and (C), leaving no residue. The oxide (C) is a liquid at room temperature and neutral to moist litmus paper while the gas (B) is a neutral oxide. White phosphorus burns in excess of (B) to produce a strong white dehydrating agent. Write balanced equations for the reactions involved in the above process.

Sol. Since the colourless inorganic salt 'A' on heating gives no residue hence it should be ammonium salt. One of the products is oxide which is liquid at room temperature and neutral. Therefore, this oxide may be water. The second product is also neutral oxide. On the basis of above discussion, the compound 'A' may be NH_4NO_3 . Its reactions are given below:



(A) (B) (C)



P_4O_{10} is strong dehydrating agent.